

Characterizing Psychiatric Comorbidities in an Alcoholic Liver Disease Cohort in a Vertically Integrated Health Care System

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Introduction

- Alcoholic liver disease (ALD) the leading cause of liver failure requiring liver transplant in the United States
- Data on prevalence and demographic profile of patients with ALD is challenging to obtain
- Alcohol use disorder and mental health conditions, especially depression and anxiety, often co-exist

Objective

We aimed to estimate prevalence and characterize a cohort of patients with alcoholic liver disease in a large, vertically integrated health care system to identify important demographic patterns and psychiatric co-morbidities.

Methods

- Historical cohort study (n = 10,655) including KPSC members ≥18 years or older from 2016-2021 with at least two instances of ICD-10 codes for ALD (validated in previous study)
- Psychiatric diagnoses were confirmed with one instance of ICD-10 code
- We calculated a crude prevalence of ALD – overall and by race/ethnicity. We approximated the denominator by using active KPSC adult members (≥18 years) in 2020.
- We completed descriptive analysis on this cohort, characterizing age, sex, race, neighborhood deprivation index (NDI), education, and income

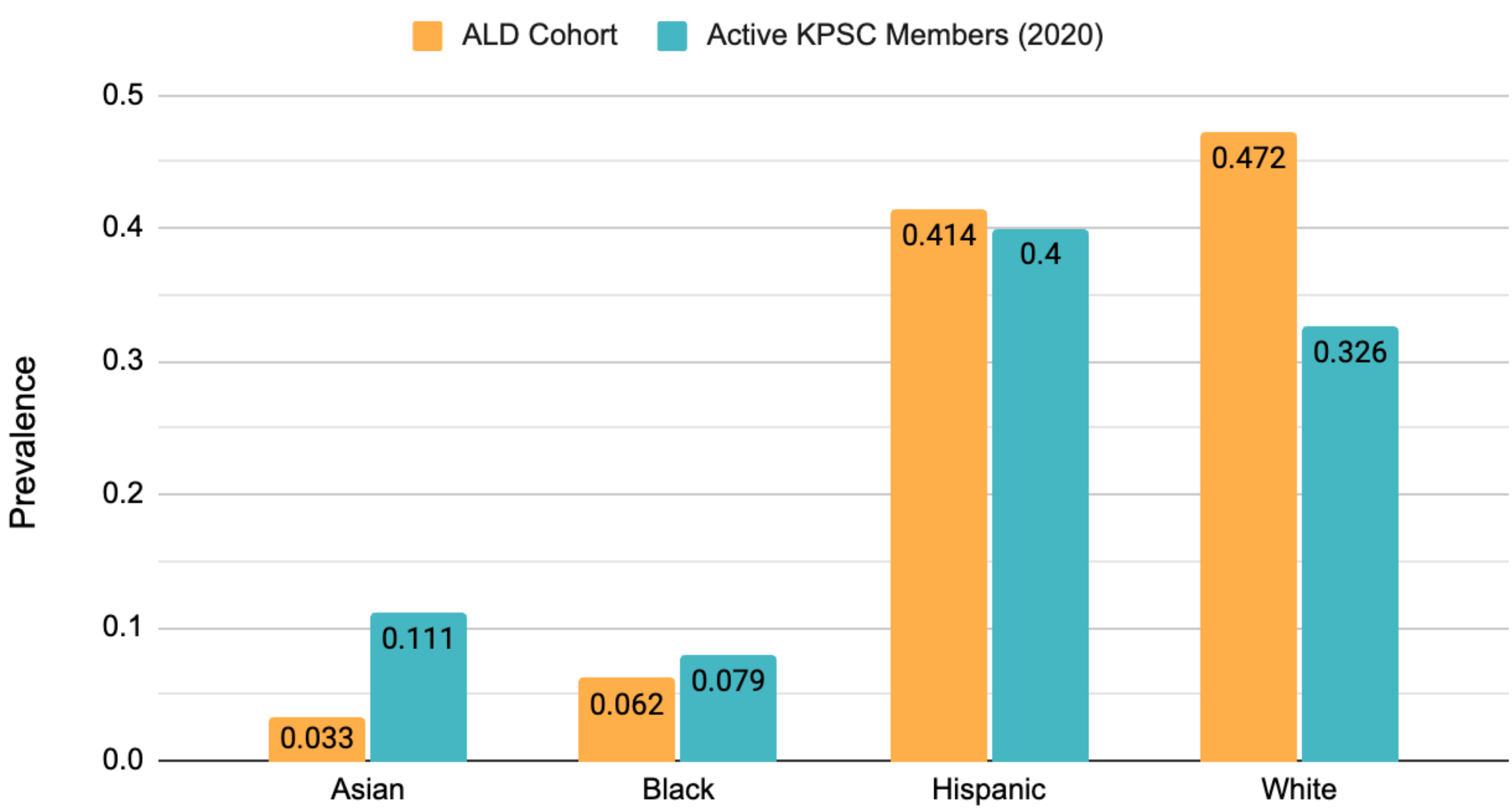


Figure 1. Prevalence of alcoholic liver disease (ALD) by race, compared with the racial distribution of the 2020 KPSC population.

Results

- We identified 10,655 patients with ALD – mean age of 58.3 years; 28.8% female; 3.3% Asian, 6.2% Black, 41.4% Hispanic, 47.2% White; 52% with at least one diagnosed psychiatric condition.
- The crude prevalence of ALD is 286.44 (281.01, 291.87) per 100,000 people.
- The majority of patients who did not have a diagnosed psychiatric condition were male, Hispanic, or had a median household income between \$50,000 - \$75,000.
- Females represent 29% of the total ALD population, 38% of patients with a psychiatric comorbidity and 19% of patients without a psychiatric comorbidity.
- Differences in mean NDI between patients with (0.2) and without a psychiatric diagnosis (0.3) are clinically insignificant.

Conclusion & Implications

- This study adds to the existing literature by reporting the prevalence of ALD in a large healthcare system.
- Female patients with ALD are overrepresented in the group of patients with a psychiatric diagnosis.
- Demographic trends in psychiatric comorbidities may imply underdiagnosis in certain populations, as opposed to true decreased prevalence.
- One limitation includes reliance on two ICD-10 codes to identify patients with ALD.
- Future work includes identifying patterns with specific psychiatric conditions and identifying variables associated with psychiatric diagnosis with a linear regression.

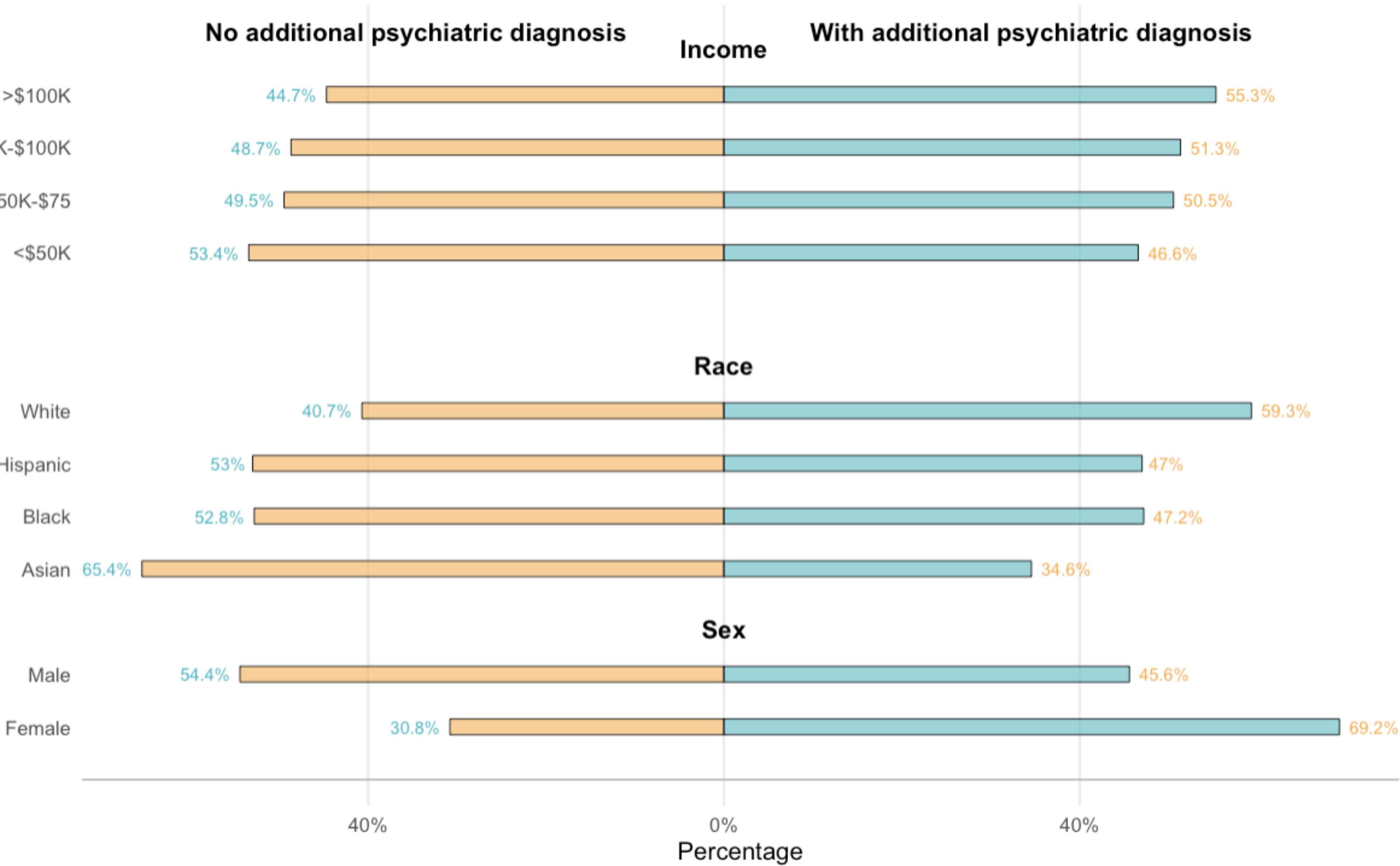


Figure 2. Diverging plot illustrating demographic differences in ALD patients with and without an additional psychiatric diagnosis.